GSE 518
Prof. Carlos A. Flores
Cal Poly State University

Problem Set 7

You must turn in problems: 1, 2, 4 and 5. However, you are strongly encouraged to work on all the problems before the final exam, since the material they cover will be part of it. Please let me know if you have any questions.

1. Let $X_1, X_2, \ldots, X_n$ and $Y_1, Y_2, \ldots, Y_m$ be two independent random samples from two distributions (populations) with means and variances (μ_x, σ_x^2) and $\left(\mu_y, \sigma_y^2\right)$, respectively. Prove that: $E\left(\overline{X} - \overline{Y}\right) = \mu_x - \mu_y$ and $Var\left(\overline{X} - \overline{Y}\right) = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}$.

2. Let $X_1, X_2, \ldots, X_n$ be a random sample. Prove that $Cov\left(X_i - \overline{X}, \overline{X}\right) = 0$ for $i = 1, \ldots, n$.

3. Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution $N(\mu, \sigma^2)$. Let $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$. Show that $Var(s^2) = 2\sigma^4/(n-1)$. (*Hint:* use Theorem 5.3.1 part (c). You can take as given that the variance of a Chi-squared with r degrees of freedom equals $2r$.)

4. Let $X_1, X_2, \ldots, X_n$ and $Y_1, Y_2, \ldots, Y_m$ be two independent random samples from normal distributions $N(\mu_x, \sigma_x^2)$ and $N\left(\mu_y, \sigma_y^2\right)$, respectively. Show that $\left[s_x^2\sigma_y^2\right]\left[s_y^2\sigma_x^2\right]^{-1}$ is distributed $F_{n-1,m-1}$. (*Hint:* use "Result 2" from class).

5. Problem 5.16 in CB.

5.15 Establish the following recursion relations for means and variances. Let $\bar{X}_n$ and S_n^2 be the mean and variance, respectively, of $X_1, \ldots, X_n$. Then suppose another observation, X_{n+1}, becomes available. Show that

(a) $\bar{X}_{n+1} = \frac{X_{n+1} + n\bar{X}_n}{n+1}$.

(b) $nS_{n+1}^2 = (n-1)S_n^2 + \left(\frac{n}{n+1}\right)(X_{n+1} - \bar{X}_n)^2$.

5.16 Let $X_i, i = 1, 2, 3$, be independent with $\mathrm{n}(i, i^2)$ distributions. For each of the following situations, use the X_is to construct a statistic with the indicated distribution.

(a) chi squared with 3 degrees of freedom

(b) t distribution with 2 degrees of freedom

(c) F distribution with 1 and 2 degrees of freedom

5.17 Let X be a random variable with an $F_{p,q}$ distribution.

(a) Derive the pdf of X.

(b) Derive the mean and variance of X.

(c) Show that $1/X$ has an $F_{q,p}$ distribution.

(d) Show that $(p/q)X/[1 + (p/q)X]$ has a beta distribution with parameters $p/2$ and $q/2$.

5.18 Let X be a random variable with a Student's t distribution with p degrees of freedom.

(a) Derive the mean and variance of X.

(b) Show that X^2 has an F distribution with 1 and p degrees of freedom.

(c) Let $f(x|p)$ denote the pdf of X. Show that

$$\lim_{p\to\infty} f(x|p) \to \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

at each value of x, $-\infty < x < \infty$. This correctly suggests that as $p \to \infty, X$ converges in distribution to a $\mathrm{n}(0, 1)$ random variable. (*Hint*: Use Stirling's Formula.)

(d) Use the results of parts (a) and (b) to argue that, as $p \to \infty, X^2$ converges in distribution to a χ_1^2 random variable.

(e) What might you conjecture about the distributional limit, as $p \to \infty$, of $qF_{q,p}$?

5.19 (a) Prove that the χ^2 distribution is *stochastically increasing* in its degrees of freedom: that is, if $p > q$, then for any a, $P(\chi_p^2 > a) \geq P(\chi_q^2 > a)$, with strict inequality for some a.

(b) Use the results of part (a) to prove that for any ν, $kF_{k,\nu}$ is stochastically increasing in k.

(c) Show that for any k, ν, and α, $kF_{\alpha,k,\nu} > (k-1)F_{\alpha,k-1,\nu}$. (The notation $F_{\alpha,k-1,\nu}$ denotes a level-α *cutoff point*; see Section 8.3.1. Also see Miscellanea 8.5.1 and Exercise 11.15.)

5.20 (a) We can see that the t distribution is a mixture of normals using the following argument:

$$P(T_\nu \leq t) = P\left(\frac{Z}{\sqrt{\chi_\nu^2/\nu}} \leq t\right) = \int_0^\infty P\left(Z \leq t\sqrt{x}/\sqrt{\nu}\right) P\left(\chi_\nu^2 = x\right) dx,$$

where T_ν is a
Theorem of C

f_{T_ν}

a scale mixtu
(b) A similar for
mixture of chi
freedom, then

where $f_\nu(y)$ is
integral expre
(c) Verify that th

is valid for all
5.21 What is the proba
exceed the populat
5.22 Let X and Y be ii
$Z^2 \sim \chi_1^2$.
5.23 Let $U_i, i = 1, 2, \ldots$
distribution

where $c = 1/(e-1$

(*Hint*: Note that
a sample of size x.)
5.24 Let $X_1, \ldots, X_n$ be

Let $X_{(1)} < \cdots <$
independent randor
5.25 As a generalization

Let $X_{(1)} < \cdots <$
$X_{(n-1)}/X_{(n)}$, and
bution of each of th